

$$01) \int_0^1 \frac{16n}{8n^2+2} dn$$

Solution:

$$\text{Let, } u = 8n^2 + 2 \checkmark$$

$$\frac{du}{dn} = 16n \checkmark$$

$$du = 16n \cdot dn \checkmark_{\text{Impotol}}$$

$$\boxed{\frac{du}{16n} = dn} \checkmark$$

⇒ Putting upper and lower limit into, $u = 8n^2 + 2$.

$$n=0 \rightarrow 2 \rightarrow \text{Lower}$$

$$n=1 \rightarrow 10 \rightarrow \text{Upper.}$$

$$= \int_2^{10} \frac{1 \cdot du}{u} = \left[\ln |u| + C \right]_2^{10} \rightarrow \text{Formula.}$$

⇒ Now Putting upper & lower limit

$$\ln |10| - \ln |2|$$

$$2.302$$

$$0.693$$

$$= \boxed{1.609} \text{ Answer}$$

$$02) \int \frac{n^2}{n^2+1} dn.$$

of
Degree Numerator > Denominator = Long Division
Degree of Numerator = Denominator = Long Division

Solution:

$$\begin{array}{r} n^2+1 \overline{) n^2} \quad (1 \rightarrow \text{Quotient.}) \\ \underline{+ n^2+1} \\ -1 \end{array}$$

-1 Remainder

Remainder
~~Quotient~~ -/+ Quotient
Denominator
Same

$$\Rightarrow \int \left(1 - \frac{1}{n^2+1} \right) dn = \int 1 dn - \int \frac{1}{n^2+1} dn.$$

$$\therefore \int \frac{1}{n^2+a^2} dn = \frac{1}{a} \cdot \tan^{-1} \left(\frac{n}{a} \right) + C.$$

$$n - \frac{1}{a} \cdot \tan^{-1} \frac{n}{a}$$

$$\boxed{n - \tan^{-1} n + C} \text{ Answer}$$

$$03) \int (\sec n - \tan n)^2 dn$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

$$= \sec^2 n - 2(\sec n)(\tan n) + \tan^2 n$$

$$\int \sec^2 n - 2 \sec n \tan n + \tan^2 n dn$$

Identity

$$\therefore \tan^2 n = \sec^2 n - 1$$

$$\int \sec^2 n - 2 \sec n \tan n + (\sec^2 n - 1) dn$$

$$= \int (2 \sec^2 n - 2 \sec n \tan n - 1) dn$$

~~2 tan n~~ Seprate use integrals of each term.

$$2 \int \sec^2 n dn - 2 \int \sec n \tan n dn - \int 1 dn$$

$$2 \tan n - 2 \sec n - n + C$$

$$= 2 \tan n - 2 \sec n - n + C \quad \text{Answer:}$$

$$04) \int_{\pi/4}^{\pi/3} \frac{dn}{\cos^2 n \tan n}$$

Solution:-

$$\int \frac{\sec^2 n}{\tan n} dn$$

$$\text{Let } u = \tan n \quad \checkmark$$

$$du = \sec^2 n dn \quad \checkmark$$

Putting upper and lower limit into u

$$u = \tan n$$

$$\text{upper} \Rightarrow \tan 60 = 1.732$$

$$\text{lower} \Rightarrow \tan 45 = 1$$

$$\int_1^{1.732} \frac{du}{u}$$

$$\therefore \frac{1}{u} dn = \ln |u| + C \quad \checkmark \text{ Formula.}$$

$$= \left[\ln |u| + C \right]_1^{1.732}$$

Putting Limit

$$\text{upper} \Rightarrow \ln |1.732|, \text{ lower} = \ln |1| =$$

$$= 0.549$$

$$\text{upper} - \text{lower} = 0.549 \quad \text{Ans}$$

$$05) \int \frac{1-n}{\sqrt{1-n^2}} dn$$

$$\therefore \frac{1}{\sqrt{a^2-n^2}} = \sin^{-1}\left(\frac{n}{a}\right) + C$$

Solution:-

Separately Integrals: ↓

$$\int \frac{1-n}{\sqrt{1-n^2}} dn = \int \frac{1}{\sqrt{1-n^2}} dn - \int \frac{n}{\sqrt{1-n^2}} dn$$

$$\text{Let, } u = 1-n^2 \checkmark$$

$$\frac{du}{dn} = -2n$$

$$du = -2n dn$$

$$\boxed{\frac{du}{-2} = n dn} \checkmark$$

$$\sin^{-1}\left(\frac{n}{1}\right) + C_1 - \int \frac{du}{\sqrt{1-n^2}}$$

$$\sin^{-1}\left(\frac{n}{1}\right) + C_1 - \frac{1}{-2} \int \frac{du}{\sqrt{1-n^2}}$$

$$\sin^{-1}\left(\frac{n}{1}\right) + C_1 - \frac{1}{2} \int \frac{1}{\sqrt{1-n^2}} du$$

$$= \sin^{-1} n - \frac{1}{2} \cdot \frac{\sqrt{4}}{1/2}$$

$$\leftarrow \sin^{-1}\left(\frac{n}{1}\right) + C_1 - \frac{1}{2} \ln|\sqrt{1-n^2}| + C_2$$

$$\sin^{-1} - \frac{1}{2} \cdot 2\sqrt{4}$$

$$\boxed{\sin^{-1} + \sqrt{u} + C} \text{ Answer}$$

$$06) \int \frac{dn}{n-\sqrt{n}}$$

$$\int \frac{dn}{\sqrt{n}\sqrt{n}-\sqrt{n}} = \int \frac{dn}{\sqrt{n}(\sqrt{n}-1)}$$

Multiple and Divide by 2 ↑

$$\frac{1}{2} \int \frac{2 dn}{\sqrt{n} (\sqrt{n}-1)}$$

$$\text{Let, } u = \sqrt{n} \checkmark$$

$$\frac{du}{dn} = \frac{1}{2} n^{-1/2}$$

$$du = \frac{1}{2} n^{-1/2} \cdot dn$$

$$\boxed{du = \frac{1}{2\sqrt{n}} dn} \checkmark$$

$$07) \int \frac{e^{-\cot z}}{\sin^2 z} dz$$

Solution:-

$$\text{Let, } u = -\cot z$$

$$du = -(\operatorname{cosec}^2 z) dz$$

$$du = +\operatorname{cosec}^2 z dz$$

$$du = \frac{1}{\sin^2 z} dz$$

$$\Rightarrow e^u \cdot du \quad \therefore e^u = e^u + C \checkmark$$

$$\boxed{e^u + C} \text{ Answer}$$

Replacing u

$$\boxed{e^{-\cot z} + C} \text{ Answer}$$

$$08) \int \frac{2^{\ln z^3}}{16z} dz$$

Solution:

$$\text{Let, } u = \ln z^3 = \boxed{3 \ln z} \checkmark$$

$$du = \left| 3 \cdot \frac{1}{z} dz \right| \checkmark$$

$$du = \frac{3}{z} dz$$

$$= \int \frac{2^u}{16z} dz$$

$$\frac{1}{16} \int \frac{2^u}{z} dz$$

$$\frac{1}{3 \times 16} \int 2^u \cdot du$$

$$= \frac{1}{48} \int 2^u \cdot du$$

$$\frac{1}{48} \left(\frac{2^u}{\ln 2} + C \right) \text{ Answer}$$

$$09) \int \frac{dz}{e^z + e^{-z}} \rightarrow \text{let, } u = e^z \checkmark, \quad \boxed{du = e^z \cdot dz} \checkmark$$

Simplify Denominator:-

$$\frac{e^z + \frac{1}{e^z}}{1} = \frac{e^{2z} + 1}{e^z}$$

$$\text{Complete Term, } \int \frac{1 dz}{\frac{e^{2z} + 1}{e^z}}$$

$$\int \frac{e^z dz}{e^{2z} + 1}$$

$$= \text{Replace } \int \frac{du}{u^2 + 1}$$

$$\frac{1}{1} \cdot \tan^{-1} \left(\frac{u}{1} \right) + C$$

$$\tan^{-1}(u) + C$$

Replacing u

$$\boxed{\tan^{-1}(e^z) + C} \checkmark$$

Answer

$$\therefore \frac{1}{a^2 + u^2} du = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$$

$$10) \int_1^2 \frac{8dn}{n^2-2n+2}$$

Solution:-

$$\int_1^2 \frac{8dn}{n^2-2n+1+1} \Rightarrow 8 \int_1^2 \frac{dn}{(n-1)^2+1} \quad \begin{array}{l} \Rightarrow \text{Putting Limit} \\ 2 \rightarrow 1 \\ 1 \rightarrow 0 \end{array}$$

$$\text{Let, } u = (n-1) \checkmark \\ du = 1dn \checkmark$$

Replacing...

$$8 \cdot \int_0^1 \frac{du}{(u)^2+1}$$

$$\therefore \frac{1}{a^2+u^2} = \frac{1}{a} \cdot \tan^{-1}\left(\frac{u}{a}\right) + C.$$

$$\Rightarrow 8 \cdot \frac{1}{u} \cdot \tan^{-1}\left(\frac{1}{u}\right) + C$$

$$8 \left[\tan^{-1}\left(\frac{1}{u}\right) + C \right]_0^1$$

Now Putting Limit

$$\tan^{-1}(1) - \tan^{-1}(0)$$

$$45^\circ - 0$$

$$8 \cdot 45^\circ = \boxed{360 \text{ OR } 2\pi} \text{ Answe}$$

$$11) \int_{-1}^0 \frac{4dn}{1+(2n+1)^2}$$

Solution:-

$$\text{Let, } u = (2n+1) \quad \begin{array}{l} \text{Putting limits in } u \\ u = (2n+1) \end{array}$$

$$du = 2dn$$

$$\begin{array}{l} 0 \rightarrow 1 \\ -1 \rightarrow -1 \end{array}$$

$$= \int_{-1}^1 \frac{(2) \cdot 2 dn}{1+(2n+1)^2} = 2 \int_{-1}^1 \frac{2dn}{1+(2n+1)^2} = 2 \int_{-1}^1 \frac{du}{1^2+(u)^2}$$

$$\therefore \int \frac{1}{a^2+u^2} = \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) + C.$$

$$= 2 \cdot \left[\frac{1}{1} \cdot \tan^{-1}(u) + C \right]$$

$$2 \cdot \left[\tan^{-1}(u) + C \right] = 2 \tan^{-1}(u) + C$$

$$2 \tan^{-1}(2n+1) + C$$

$$\text{Put lower} \\ = \cancel{180} - 90$$

$$\text{Put upper} \\ = \cancel{180} 90$$

$$90 - (-90) = \boxed{180} \text{ Answe}$$

$$12) \int_{-1}^3 \frac{4n^2 - 7}{2n + 3} dn$$

Solution:-

Degree of Num > Denom = Long Division.

$$\begin{array}{r} 2n+3 \overline{) 4n^2-7} \quad (2n-3) \text{ quotient} \\ \underline{+ 4n^2+6n} \\ -6n-7 \\ \underline{+ 6n+9} \\ 2 \end{array}$$

(2) → Remainder

$$= \int_{-1}^3 \left(2n - 3 + \frac{2}{2n+3} \right) dn$$

$$= \int_{-1}^3 2n - 3 dn + \int \frac{2}{2n+3} dn$$

$$\frac{2 \cdot n^2 - 3n}{2}$$

let $u = 2n+3$
 $du = 2dn$

$$n^2 - 3n + \int \frac{du}{u}$$

$$n^2 - 3n + \int \frac{1 du}{u}$$

$$\therefore \ln|u| + C$$

$$n^2 - 3n + \left[\ln|u| + C \right]_{-1}^3$$

Putting limit on u
 $u = 2n+3$

$$\left[\frac{n^2 - 3n}{2} \right]_{-1}^3 + \left[\ln|u| + C \right]_0^{2.197}$$

upper = 2.197
 lower = 0

$$9 - 3/2$$

$$9 - 1.5$$

$$0 \rightarrow \text{upper} + \ln|u| + C$$

$$= 0 - 4$$

$$-4 + 2.197$$

$$= -1.803$$

$$13) \int \frac{dt}{1 - \sec t}$$

$$\frac{\sin^2 n + \cos^2 n}{\cos^2 n} = \frac{1}{\cos^2 n}$$

$$\tan^2 n + 1 = \sec^2 n$$

$$\tan^2 n = \sec^2 n - 1$$

$$-\tan^2 n = 1 - \sec^2 n$$

Solution:-

Rationalize.

$$\frac{1}{1 - \sec t} \cdot \frac{1 + \sec t}{1 + \sec t} \Rightarrow \frac{1 + \sec t}{1 - \sec^2 t}$$

$$\therefore \boxed{-\tan^2 t = 1 - \sec^2 t}$$

Replacing--

$$= \frac{1 + \sec t}{-\tan^2 t} = -\frac{1}{\tan^2 t} - \frac{\sec t}{\tan^2 t}$$

$$= -\cot^2 t - \frac{1}{\cos t \cdot \frac{\sin^2 t}{\cos^2 t}} = -\cot^2 t - \frac{1}{\frac{\sin^2 t}{\cos t}}$$

$$-\cot^2 t - \frac{\cos t}{(\sin t)^2}$$

identity
 $\therefore -\cot^2 t = 1 - \operatorname{cosec}^2 t$

$$= \int \left(1 - \operatorname{cosec}^2 t - \frac{\cos t}{\sin^2 t} \right) dt$$

$$\int 1 dt - \int \operatorname{cosec}^2 t dt - \int \frac{\cos t}{\sin^2 t} dt$$

$$\text{Let, } u = \sin t$$

$$du = \cos t dt$$

$$= t - (-\cot t) - \int \frac{du}{u^2}$$

$$t + \cot t - \int u^{-2} du$$

$$t + \cot t - \frac{u^{-2+1}}{-2+1} + C$$

$$t + \cot t - \frac{u^{-1}}{-1} + C$$

$$t + \cot t + \frac{1}{u} + C$$

Replace u

$$\boxed{t + \cot t + \frac{1}{\sin t} + C}$$

$$14) \int \operatorname{cosec} t \sin 3t \, dt$$

Solution:

Rewriting: $\frac{1}{\sin t} \cdot \sin 3t \Rightarrow \frac{\sin 3t}{\sin t}$

$$\int \operatorname{cosec} t \cdot \sin 3t \, dt \equiv \int \frac{\sin 3t}{\sin t} \, dt$$

$$\equiv \sin 3t \Rightarrow \sin(2t + t)$$

$$\downarrow \quad \therefore \sin(\alpha + \beta)$$

$$= \frac{\sin 2t \cos t + \cos 2t \sin t}{\sin t} \equiv \sin 2t \cos t + \cos 2t \sin t$$

$$\int \left(\frac{\sin 2t \cos t}{\sin t} + \frac{\cos 2t \sin t}{\sin t} \right) dt$$

$$\therefore \sin 2t = 2 \sin t \cos t$$

$$\int \left(\frac{2 \sin t \cos t \cdot \cos t}{\sin t} + \frac{\cos 2t \sin t}{\sin t} \right) dt$$

$$\int \frac{2 \cancel{\sin t} \cos t \cdot \cos t}{\cancel{\sin t}} \, dt + \int \frac{\cos 2t \cdot \cancel{\sin t}}{\cancel{\sin t}} \, dt$$

Identify $\therefore \cos^2 t = 1 - \sin^2 t$

$$\int 2 \cos^2 t \, dt + \int \cos 2t \, dt$$

$$\hookrightarrow \int 2(1 - \sin^2 t) \, dt$$

$$15) \int_0^{\pi/4} \frac{1 + \sin \theta}{\cos^2 \theta} d\theta$$

Solution:-

$$= \int_0^{\pi/4} \frac{1}{\cos^2 \theta} d\theta + \int_0^{\pi/4} \frac{\sin \theta}{\cos^2 \theta} d\theta$$

$$= \int_0^{\pi/4} \sec^2 \theta d\theta + \int_0^{\pi/4} \frac{1}{\cos \theta} \cdot \frac{\sin \theta}{\cos \theta} d\theta$$

$$\left[\tan \theta \right]_0^{\pi/4} + \left[\sec \theta \cdot \tan \theta \right]_0^{\pi/4}$$

Putting upper & lower limit in each of above terms.

$$\left[\tan \pi/4 - \tan 0 \right] + \left[(\sec \pi/4 \cdot \tan \pi/4) - (\sec 0 \cdot \tan 0) \right]$$

$$\left[1 - 0 \right] + \left[(1.414 \cdot 1) - (1 \cdot 0) \right]$$

$$1 + 1.414$$

$$\boxed{2.414} \text{ Answer}$$

$$16) \int \frac{d\theta}{\sqrt{20 - \theta^2}}$$

Solution:

$$\text{Simplify: } \int \frac{d\theta}{-(\theta^2 - 20)} = \frac{1}{(-(\theta^2 - 20 + 1) - 1)}$$

$$= \frac{1}{\sqrt{-(\theta^2 - 1)^2 + 1}} = \frac{1}{\sqrt{1 - (\theta^2 - 1)^2}}$$

$$= \int \frac{1}{\sqrt{1 - (\theta^2 - 1)^2}} d\theta \text{ Simplified. } \checkmark$$

$$\therefore \frac{1}{\sqrt{a^2 - u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + C$$

$$\text{Let, } u = \theta - 1$$

$$\boxed{du = 1 d\theta}$$

$$\int \frac{du}{1 - u^2}$$

$$\sin^{-1} \left(\frac{u}{1} \right) + C \checkmark \text{ Answer}$$

$$17) \int \frac{\ln y}{y + 4y \ln^2 y} dy$$

Solution:-

$$\int \frac{\ln y}{y + 4y \ln^2 y} dy = \frac{\ln y \, dy}{y(1 + 4(\ln^2 y))}$$

$$\left\{ \begin{array}{l} \text{Let, } u = \ln y \\ du = \frac{1}{y} dy \end{array} \right.$$

$$= \frac{du}{(1 + 4(\ln y)^2)}$$

$$\frac{du}{1 + 4 \cdot (u)^2} = \frac{du}{1 + 4u^2}$$

$$= \frac{u du}{1 + 4u^2}$$

$$\text{Let } v = 1 + 4u^2$$

$$\frac{dv}{du} = 8u \Rightarrow dv = 8u \cdot du$$

$$\frac{dv}{8} = u \cdot du$$

$$\int \frac{dv}{8} = \frac{1}{8} \int \frac{dv}{v}$$

$$\frac{1}{8} \ln |v| + C$$

$$\frac{1}{8} \ln |1 + 4u^2| + C$$

$$18) \int \frac{2\sqrt{y}}{2\sqrt{y}} dy$$

$$\text{Let } u = \sqrt{y}$$

$$\frac{du}{dy} = \frac{1}{2} y^{-1/2}$$

$$du = \frac{1}{2\sqrt{y}} dy$$

$$\boxed{du \cdot 2\sqrt{y} = dy}$$

$$\Rightarrow \frac{2^4 \cdot du \cdot \cancel{2\sqrt{y}}}{2\sqrt{y}}$$

$$= \int 2^4 \cdot du$$

$$\boxed{\frac{a^n}{\ln|a|} + C}$$

Answer

Replaces

$$\frac{2^4}{\ln|2|} + C \checkmark$$

$$19) \int \frac{d\theta}{\sec \theta + \tan \theta}$$

Solution:

$$\frac{1}{\sec \theta + \tan \theta}$$

$$\frac{\sec \theta - \tan \theta}{\sec \theta - \tan \theta}$$

$$= \frac{\sec \theta - \tan \theta}{\sec^2 \theta - \tan^2 \theta}$$

$$= \int \frac{\sec \theta - \tan \theta}{1} d\theta$$

$$\because \sec^2 \theta - \tan^2 \theta = 1$$

identity

$$= \int \sec \theta d\theta - \int \tan \theta d\theta$$

Again

$$\int \frac{d\theta}{\sec \theta + \tan \theta}$$

$$= \frac{d\theta}{\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}}$$

$$\Rightarrow \frac{d\theta}{\frac{1 + \sin \theta}{\cos \theta}}$$

$$\frac{d\theta}{\frac{1 + \sin \theta}{\cos \theta}}$$

$$= \int \frac{\cos \theta}{1 + \sin \theta} d\theta$$

$$\text{Let, } u = 1 + \sin \theta$$

$$du = \cos \theta d\theta$$

$$\int \frac{du}{u} = \int \frac{1 du}{u}$$

$$\therefore \frac{1 du}{u} = \ln|u|$$

$$= \ln|u| + C$$

$$\text{replace } u \Rightarrow \ln|1 + \sin \theta| + C \text{ Answer}$$

Q1) $\int \frac{4t^3 - t^2 + 16t}{t^2 + 4} dt$ $\Rightarrow$ Degree of Num > Deno = Long Divison | Degree of Num = Deno = Long Div

$$\begin{array}{r} t^2 + 4 \overline{) 4t^3 - t^2 + 16t} \quad (4t - 1) \\ \underline{+ 4t^3 + 16t} \\ -t^2 \end{array}$$

$$\begin{array}{r} -t^2 \\ \underline{+ t^2 + 4} \\ +4 \end{array} \rightarrow \text{Remainder}$$

Quotient = $4t - 1$
 $4t - 1$

$$\Rightarrow \int \left(4t - 1 + \frac{4}{t^2 + 4} \right) dt$$

$$\int 4t dt - \int 1 dt + \int \frac{4}{t^2 + 4} dt$$

$$\int 4t dt - \int 1 dt + 4 \cdot \int \frac{1}{t^2 + (2)^2} dt$$

$$\frac{4t^2}{2} - t + 4 \cdot \frac{1}{2} \tan^{-1} \left(\frac{t}{2} \right) + C$$

$\therefore \frac{1}{a} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right) + C$

$$2t^2 - t + 2 \cdot \tan^{-1} \left(\frac{t}{2} \right) + C$$

Answer

Q2) $\int \frac{n+2}{2n\sqrt{n-1}} dn$

$$\frac{n}{2n\sqrt{n-1}} + \frac{2\sqrt{n-1}}{2n\sqrt{n-1}}$$

$$= \int \frac{1}{2\sqrt{n-1}} + \frac{1}{n} dn \Rightarrow \int \frac{1}{2\sqrt{n-1}} dn + \int \frac{1}{n} dn$$

Let, $u = \sqrt{n-1}$
 $du = \frac{1}{2\sqrt{n-1}} dn$

$$\int du + \ln|u| + C$$

$$u + \ln|u| + C \Rightarrow \sqrt{n-1} + \ln|\sqrt{n-1}| + C$$

$$23) \int_0^{\pi/2} \sqrt{1 - \cos \theta} \, d\theta$$

Solution:

$$\text{Rationalize it} \Rightarrow \frac{\sqrt{1 - \cos \theta}}{1} \cdot \frac{\sqrt{1 + \cos \theta}}{\sqrt{1 + \cos \theta}}$$

$$\therefore \frac{1^2 - \cos^2 \theta = \sin^2 \theta}{\text{identity}} = \frac{\sqrt{1^2 - \cos^2 \theta}}{\sqrt{1 + \cos \theta}} \Rightarrow \frac{\sqrt{\sin^2 \theta}}{\sqrt{1 + \cos \theta}} \Rightarrow \frac{\sin \theta}{\sqrt{1 + \cos \theta}}$$

$$= \frac{\sin \theta}{\sqrt{1 + \cos \theta}} \Rightarrow \text{Now Multiple \& Divide (2)} \\ -2 \int_0^{\pi/2} \frac{-\sin \theta}{2 \sqrt{1 + \cos \theta}} \, d\theta$$

Substitution:-

$$\text{Let, } u = \sqrt{1 + \cos \theta} \cdot 1 + \cos \theta$$

$$du = -\sin \theta \, d\theta$$

$$\int_0^{\pi/2} \frac{-\sin \theta}{\sqrt{1 + \cos \theta}} \, d\theta \Rightarrow \frac{du}{\sqrt{u}} \quad \text{Replacing}$$

Putting Limit:-

$$u = 1 + \cos \theta$$

$$\text{lower} = 2$$

$$\text{upper} = 1$$

$$\text{upper} - \text{lower}$$

$$1 - 2$$

$$\boxed{-1}$$

$$\left[\frac{du}{\sqrt{u}} \right]_2^1$$

$$\ln |\sqrt{u}| + C$$

$$24) \int (\sec t + \cot t)^2 dt$$

Solution:- $(a+b)^2 = a^2 + 2ab + b^2$

$$\sec^2 t + 2(\sec t)(\cot t) + (\cot^2 t)$$

$$\sec^2 t + 2 \sec t \cot t + \cot^2 t$$

$$\sec^2 t + 2 \cdot \frac{1}{\cos t} \cdot \frac{\cos t}{\sin t} + \cot^2 t \Rightarrow \sec^2 t + \frac{2}{\sin t} + \cot^2 t$$

$$\int (\sec^2 t + 2 \operatorname{cosec} t + \cot^2 t) dt$$

$\therefore \frac{1}{\sin t} = \operatorname{cosec} t$

$$\int \sec^2 t dt + \int 2 \operatorname{cosec} t dt + \int \cot^2 t dt - \int 1 dt$$

Identity,
 $\cot^2 t = \operatorname{cosec}^2 t - 1$

$$\int \tan t + 2 \cdot (\operatorname{cosec} t) + \text{Formula?} (15)$$

SEE

$$\tan t - 2 \ln |\operatorname{cosec} t + \cot t| - \cot t - t$$

$$\tan t - 2 \ln |\operatorname{cosec} t + \cot t| - \cot t - t + C$$

$$25) \int \frac{1}{\sqrt{e^{2y} - 1}} dy \Rightarrow \text{simply write}$$

$$\frac{1}{\sqrt{(e^y)^2 - 1}} dy$$

Let, $u = e^y$, $du = e^y dy$

$$\frac{du}{e^y} = dy$$

$$= \frac{dy}{\sqrt{(e^y)^2 - 1}} = \frac{du}{u \sqrt{u^2 - 1}}$$

$$25) \int \frac{dy}{\sqrt{e^{2y}-1}} = \int \frac{dy}{(e^y)^2-1}$$

Solution:-

$$\text{Let, } u = e^y$$

$$\frac{du}{dy} = e^y$$

$$du = e^y dy = u$$

$$du = u dy$$

$$\frac{du}{u} = dy$$

$$\boxed{\frac{du}{u} = dy}$$

$$\Rightarrow \frac{du}{u\sqrt{u^2-1^2}}$$

$$\therefore \frac{1}{u\sqrt{u^2-a^2}}$$

$$= \frac{1}{a} \sec^{-1} \left| \frac{u}{a} \right| + C$$

$$= \frac{1}{1} \sec^{-1} \left(\frac{u}{1} \right) + C$$

Replacing (u)

$$\frac{1}{1} \sec^{-1} (e^y) + C$$

$$26) \int \frac{6 dy}{\sqrt{y}(1+y)}$$

Substitute, Let

$$u = \sqrt{y}, \quad u^2 = y$$

$$du = \frac{1}{2\sqrt{y}} dy$$

$$\boxed{\frac{du}{2\sqrt{y}} = \frac{1}{2\sqrt{y}} dy}$$

$$\therefore du = \frac{1}{2\sqrt{y}} dy$$

$$\therefore u^2 = y$$

Replacing...

$$12 \int \frac{du}{1+u^2}$$

$$12 \cdot \frac{1}{1} \tan^{-1} \left(\frac{u}{1} \right) + C$$

$$\boxed{12 \tan^{-1} \cdot u + C}$$

Replacing "u"

$$\boxed{12 \tan^{-1} \cdot \sqrt{y} + C} \quad \text{A}$$

$$\int \frac{6 dy}{\sqrt{y}(1+y)}$$

Multiply and divide by 2

$$6 \cdot 2 \cdot \int \frac{1 \cdot dy}{2\sqrt{y}(1+y)}$$

$$12 \int \frac{1 dy}{2\sqrt{y}(1+y)}$$

$$\therefore \frac{1}{a^2+u^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$$

$$27) \int \frac{2dn}{n\sqrt{1-4ln^2}}$$

Solution:

Rewrite: $\int \frac{2dn}{n\sqrt{1-(2ln)^2}}$

Substitute:

Let, $u = 2ln$ ✓
 $du = \frac{2}{n} dn$ ✓

Now Substituting:

$$= \int \frac{du}{\sqrt{1-u^2}}$$

We have formula:-

$$\therefore \frac{1}{\sqrt{a^2-n^2}} = \sin^{-1}\left(\frac{n}{a}\right) + C$$

using
Formula

$$= \sin^{-1}\left(\frac{u}{1}\right) + C$$

Replacing "u"

$$\sin^{-1} \cdot u + C$$

$$\boxed{\sin^{-1} 2ln + C} \text{ Answer}$$

$$28) \int \frac{dn}{(n-2)\sqrt{n^2-4n+3}}$$

Solution:-

Rewrite: $\int \frac{1}{(n-2)\sqrt{n^2-4n+3+1-1}} = \int \frac{1 dn}{(n-2)\sqrt{(n-2)^2-1^2}}$

$(n-2)^2 \equiv n^2-4n+4$

Substitute:

Let, $u = n-2$ $\Rightarrow \frac{du}{dn} = 1$
 $du = 1dn$

$$= \int \frac{du}{u\sqrt{u^2-1^2}}$$

Formula: $\frac{dn}{n\sqrt{n^2-a^2}} = \frac{1}{a} \sec^{-1}\left|\frac{n}{a}\right| + C$

Expanding Formula:-

$$= \frac{1}{1} \sec^{-1}\left|\frac{u}{1}\right| + C$$

$$\boxed{\sec^{-1}(n-2) + C}$$

Answer
✓

$$29) (\operatorname{cosec} n - \sec n) (\sin n + \cos n) \, dn$$

Multiplying... $\rightarrow$

$$\operatorname{cosec} n \sin n + \operatorname{cosec} n \cos n - \sec n \sin n - \sec n \cos n$$

$$\frac{1}{\sin n} \cdot \sin n + \frac{1}{\sin n} \cdot \cos n - \frac{1}{\cos n} \cdot \sin n - \frac{1}{\cos n} \cdot \cos n$$

$$\cancel{1} + \frac{1 \cdot \cos n}{\sin n} - \frac{\sin n}{\cos n} - \cancel{1}$$

$$= \frac{\cos n}{\sin n} - \frac{\sin n}{\cos n}$$

$$= \int \cot n - \tan n \, dn$$

Separately Applying Integration:

$$\int \cot n \, dn + \int -\tan n \, dn$$

$$\int \frac{\cos n}{\sin n} \, dn + \int -\frac{\sin n}{\cos n} \, dn$$

Let Substitutes:

$$u = \sin n$$

$$du = \cos n \, dn$$

Substituting $\rightarrow$

Let, Substitute:

$$u = \cos n$$

$$du = -\sin n \, dn$$

$\sin n$

$$\int \frac{du}{u} + \int \frac{-du}{u}$$

$$\therefore \frac{1}{u} = \boxed{\ln|u| + C} \text{ same apply formula on } b/s$$

$$-\ln|u| + C_1 + \ln|u| + C_2$$

Now, Replacing "U" from both sides:

$$\boxed{\ln|\sin n| + C_1 + \ln|\cos n| + C_2} \text{ Ans.}$$

$$30) \int 3 \sinh\left(\frac{n}{2} + \ln 5\right) dn$$

Solution:-

$$\text{Let, } u = \left(\frac{n}{2} + \ln 5\right) \quad du = \frac{1}{2} dn = dn = 2du$$

$$\int 3 \sinh(u) 2du$$

$$3 \cdot 2 \int \sinh(u) du \Rightarrow 6 \cosh(u) + C$$

Replacing "u"

$$\boxed{6 \cosh\left(\frac{n}{2} + \ln 5\right) + C} \text{ Ans.}$$

$$31) \int_{\sqrt{2}}^3 \frac{2n^3}{n^2-1} dn \quad \text{Degree of Num} > \text{Denom} = \text{Long Division}$$

$$\begin{array}{r} n^2-1 \overline{) 2n^3} \quad \left(\begin{array}{l} 2n^3 \rightarrow \text{quotient} \\ 2n^3 - 2n \end{array} \right) \quad \left| \begin{array}{l} 2n \\ 2n^3 - 2n \end{array} \right. \\ \underline{+ 2n^3 - 2n} \quad \left(\begin{array}{l} + 2n \rightarrow \text{Rem} \end{array} \right) \end{array}$$

$$\int \frac{2n + \frac{2n}{n^2-1}}{n^2-1} dn = \int 2n dn + \int \frac{2n}{n^2-1} dn$$

$$= \left[\frac{2 \cdot n^2}{2} \right]_{\sqrt{2}}^3 + \left[\ln |n^2-1| \right]_{\sqrt{2}}^3$$

Putting limits

$$\begin{array}{l} \text{upper} = 9 \\ \text{lower} = 2 \end{array} \quad 9-2=7$$

$$\begin{array}{r} \xrightarrow{\text{upper}} \quad \quad \quad \xleftarrow{\text{lower}} \\ 7 + \ln |3^2-1| - \ln |\sqrt{2}-1| \\ 2.07 - (-0.88) \\ 2.07 + 0.88 \\ = 2.95 \end{array}$$

$$7 + 2.95 = 9.95 \approx 10 \checkmark$$